

CHARLES LANCE QUARLES

Senior Python Developer | FastAPI, Django, AWS, PostgreSQL, Docker | AI/ML Integration

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PROFESSIONAL SUMMARY

Senior Python Developer with 9 years of expertise in designing and deploying scalable backend systems and AI/ML models. Spearheaded microservices architecture using FastAPI, improving response times by 45%. Delivered consistent 99.9% uptime for RESTful APIs, accelerating feature releases by 30% for five frontend teams. Mentored a team of 4 engineers, establishing Git best practices.

PROFESSIONAL EXPERIENCE

Senior Python Developer - Team Lead

AdvSoft LLC

Philadelphia, PA

06/2021 - Present

- Led the design and implementation of a microservice based architecture using FastAPI, reducing response times by 45% across critical backend services.
- Transformed raw data into actionable insights by establishing data pipelines integrating AI/ML models with PyTorch, resulting in a 30% reduction in data processing time.
- Pioneered resilient RESTful APIs, delivering a consistent 99.9% uptime while managing peak loads, facilitating a 30% faster feature release rate.
- Built a custom authentication system using OAuth2 and JWT tokens, integrating with external identity providers like Auth0 and Okta.
- Engineered infrastructure-as-code templates via Terraform and AWS CloudFormation, creating consistent deployments across staging and production, which also enabled a 25% increase in deployment frequency.
- Directed the migration of legacy applications to a containerized environment using Docker and Kubernetes (EKS), cutting infrastructure costs by 30% through efficient resource utilization.
- Developed reusable React components with TypeScript and Tailwind CSS, contributing to a component library that reduced front-end development time by 15%.
- Orchestrated asynchronous task processing leveraging Celery and Redis, diminishing non-critical computation times by 40% and improving frontend responsiveness by 35% across the platform.
- Detected and triaged 7 critical system vulnerabilities using Prometheus and Grafana dashboards, collaborating with cross-functional teams to remediate issues before impacting production.
- Transformed the model deployment process by implementing AWS Lambda functions and SageMaker, driving a 1% increase in overall model accuracy through faster iteration cycles and model tuning.
- Led and mentored a team of 4 backend engineers; conducted code reviews, managed sprint planning, and established best practices in Git branching strategies.
- Documented APIs and internal systems using Swagger/OpenAPI and MkDocs, improving onboarding speed for new hires by 40%.

Senior Python Developer

Code Power

Farmington, MI

06/2018 - 05/2021

- Built and maintained robust Django REST APIs for handling millions of monthly data points in logistics analytics.
- Refactored legacy data system to event-driven architecture using RabbitMQ, increasing data throughput by 15,000 data points/second and reducing data latency to under 100 milliseconds.
- Directed the containerization and deployment of microservices on AWS ECS, optimizing resource utilization by 20% and supporting the processing of millions of data points.
- Developed AI-driven data cleaning and classification pipelines using Pandas, Scikit-learn, and NumPy, improving data quality by 70%.
- Orchestrated the deployment of machine learning models into production by collaborating with the AI team, utilizing Flask inference servers and cutting model deployment time by 4 weeks.
- Spearheaded the revamping of legacy admin panels with Django Admin, constructing custom filters and exports, which augmented administrator acceptance by 75% and reduced errors by 20%.
- Engineered a custom static analysis tool that automatically checks for common security flaws, achieving a 95% compliance rate with OWASP guidelines across the Django codebase.
- Re-architected database schema for stock data using MySQL, dropping API latency by 35ms, uplifting data integrity by 30%, and serving as the most junior engineer.

PROFESSIONAL EXPERIENCE

Software Developer

StartupSoft

San Francisco, CA

08/2017 - 04/2018

- Spearheaded migration of legacy API endpoints to a Django REST Framework, slashing average response time by 20% and improving application performance on 3 client-facing applications.
- Delivered enhancement of application capabilities by incorporating 3 external APIs, expanding functionality and achieved 99.99% uptime.
- Standardized data ingestion methods, resulting in a consistent 98% data accuracy rate across all AI/ML projects, and also contributed to feature release.
- Automated testing processes using Pytest and Jenkins CI, improved code quality scores by 10%, demonstrating commitment to software excellence.
- Transformed the deployment process with AWS and Docker, boosting application performance by 25% and implemented robust monitoring, improving issue detection by 60%.

Python Developer

CodingCops

Chicago, IL

08/2016 - 07/2017

- Constructed 9 automated test scripts with Python, improving code test coverage by 15%.
- Mastered Linux system administration, Git version control, and SQL database management, enabling independent identification and resolution of intermittent data pipeline incidents within two hours.
- Collaborated with senior engineers to hone skills on Linux, Bash, Git, and SQL, streamlining ETL and data cleaning protocols by 8%, exceeding initial efficiency targets.

TECHNICAL SKILLS

Frontend Technologies: HTML5, CSS3, Bootstrap, JavaScript, TypeScript, React.js, Next.js, Tailwind CSS

Backend Technologies: Python, FastAPI, Django, Flask, SQLAlchemy, API Design, Microservices Architecture

Cloud & Infrastructure: AWS (EC2, Lambda, S3, RDS, ECS), Microsoft Azure, Serverless Architecture, Docker, Kubernetes

Data Engineering & AI/ML: PyTorch, Scikit-learn, Pandas, NumPy, ETL, Data Pipelines, Machine Learning, Deep Learning

DevOps & Automation: Git, GitHub Actions, Jenkins, Terraform, CI/CD, Automated Testing

Security: OAuth2, JWT, OWASP, Authentication, Authorization

Databases: PostgreSQL, MySQL, Redis, MongoDB, NoSQL

Architecture & Protocols: REST APIs, Microservices, OAuth2, JWT, Celery, RabbitMQ, WebSockets, gRPC

Testing & Monitoring: Pytest, Unit Testing, Prometheus, Grafana, Sentry

EDUCATION

Bachelor's degree in Computer Science

Becker College

Worcester, MA

08/2012 - 06/2016